



Lightning EarPods bug

Apple has supposedly solved a recent volume control bug on Lightning EarPods. <http://dgit.in/iEarPodsBug>



Olympus at Photokina

Olympus has finally launched its OM-D E-M1 Mark II at the Photokina 2016. <http://dgit.in/OlympCams>

Space Age

One week on, investigations had still not yielded a conclusive result, and Musk expressed his frustration over the failure as he tweeted that it was "turning out to be the most difficult and complex failure we have ever had in 14 years." Musk has also belabored the minute point that the "explosion" was more accurately a very fast fire – it just appears instantaneous to the human eye. In an attempt to crowdsource some help for the investigations, he took to Twitter



16th July 1969. Perhaps the most iconic and definitive image from the advent of the Space Age, this view from the Apollo 11 spacecraft shows Earth rising above the moon's horizon.

to ask for recordings of the event to be emailed to SpaceX to aid investigations. One of the replies points out that a YouTube video indicates something hitting the rocket, in response to which Musk said that they had not ruled that out.

Currently, the investigation is being carried out by a team consisting of about 20 people from SpaceX, the Federal Aviation Administration, NASA, the US Air Force, and the aerospace industry. The agreement between SpaceX and the FAA means that the latter is only acting in an advisory role. Investigations started almost immediately after the incident, as the SpaceX team drew on their formidable arsenal of almost 3,000 sensors and cameras, reviewing a duration of several milliseconds prior to the explosion.

The repercussions: Who else will be affected, and why?

Marco Caceres, Director of Space Studies at Teal Group, a business analytics

company specializing in defense and aerospace markets, said in an interview with Bloomberg that he does not expect this failure to significantly impact future contracts for SpaceX. He speculated that the circumstances of the failure may point to the problem not entirely being with the launch vehicle itself – since the failure occurred while on the launch pad without the engines on.

Tory Bruno, CEO of United Launch Alliance, SpaceX's major industry rival, believes that the explosion will ground SpaceX for 9-12 months. Talking to Reuters, he added that the time taken in rebuilding the launch pad, which also sustained damage in the explosion, is usually not the bottleneck in such cases.

The exploding rocket also destroyed with it some precious cargo. The Falcon 9's payload was a \$200 million Israeli satellite that was commissioned in part by Facebook to provide internet access to areas of Africa and Europe. The satellite was built by Israeli Aerospace Industries (IAI) for Israeli telecom company Spacecom, who are set to receive insurance compensation from both SpaceX as well as IAI. SpaceX are offering either a \$50 million compensation or a free launch in the future, while IAI will pay over \$200 million after including the interest with the cost of the satellite. Spacecom was set to be bought by Chinese conglomerate Beijing Xinwei for \$285 million, but the deal was initially contingent on the Amos-6 being successfully deployed. However, Spacecom CEO David Pollock has indicated to SpaceNews that the deal may still be on, albeit with renegotiations over price.

The story so far, and why Elon Musk is just the man to turn things around.

In 2002, Musk, at the age of 31, was already a multi-millionaire. Instead of retiring to an idyllic life in the Bahamas, he has spent the intervening 14 years establishing the groundwork for humans becoming a multi-planetary species, starting with the aim of colonizing Mars. Why? Because he's Elon Musk. But such bold and seemingly ridiculous targets come with extremely high risk,

as the company found out once again on September 1st.

Elon Musk knows has known failure and near-failure on a grand and terrifying scale. It's almost a necessary prerequisite for being a self-made, billion-dollar-net-worth individual. Space travel is inherently a high-risk industry. The astronomical (no pun intended) odds against success of setting up a company that manufactures and launches rockets, and then lands them upright on a drone ship at sea, meant that Musk had to fund SpaceX with his private money. Investors are far too risk-averse to fund an unproven startup with the stated goal of making space travel affordable. And Musk found his considerable finances stretched right down to the wire before SpaceX could even glimpse success. SpaceX only began with enough funding for four launches. Not surprisingly, a success for SpaceX would be considered not completing production of a unit, but also successfully launching said unit into space.

Their first three launches failed. The rockets exploded before they could reach orbit. Their finances were a mess, not only because Musk had also launched Tesla Motors and SolarCity within the previous 4 years – again using personal funds – but also by the fact that the global financial crisis of 2007 crippled the automobile industry.

However, in September 2008, SpaceX's last effort at launching a rocket into orbit succeeded, and NASA stepped in and handed SpaceX a cool \$1.2 billion contract to make deliveries to the International Space Station using their rockets. SpaceX was saved at the eleventh hour, and it marked the beginning of a turnaround for Musk's other companies as well.

It's instructive to keep in mind that in having such a bold mission statement and in making tangible progress towards it, Elon Musk is in pursuit of a goal so audacious that two of the world's superpowers had a clash of egos about it. We are referring, of course, to the space race that ensued after the Cold War between the USA and the USSR. The rivalry between countries to put the first human in space – and then on the moon